



Mastering Excel Math Functions

Essential formulas for calculation and data analysis

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SUMIF: Conditional Summing

What It Does

Adds cells specified by a given condition or criteria. Perfect for targeted calculations based on specific rules.

When to Use

Sum numbers *only if* they meet your rule. For example: "Sum sales only for Apples" or "Sum scores greater than 50."



Syntax

`=SUMIF(range, criteria, [sum_range])`



Example 1

`=SUMIF(B2:B10, ">100")`

Sum values greater than 100



Example 2

`=SUMIF(A2:A10, "Apple", B2:B10)`

Sum Column B where Column A is "Apple"

ABS: Absolute Value

The Concept

Returns the absolute value of a number by converting negative numbers to positive. It removes the minus sign, giving you the distance from zero.

Common Use Cases

- Finding differences where direction doesn't matter
- Calculating distance between values
- Error margin calculations
- Data normalization

1

Syntax

`=ABS(number)`

2

Convert Negative

`=ABS(-50) → 50`

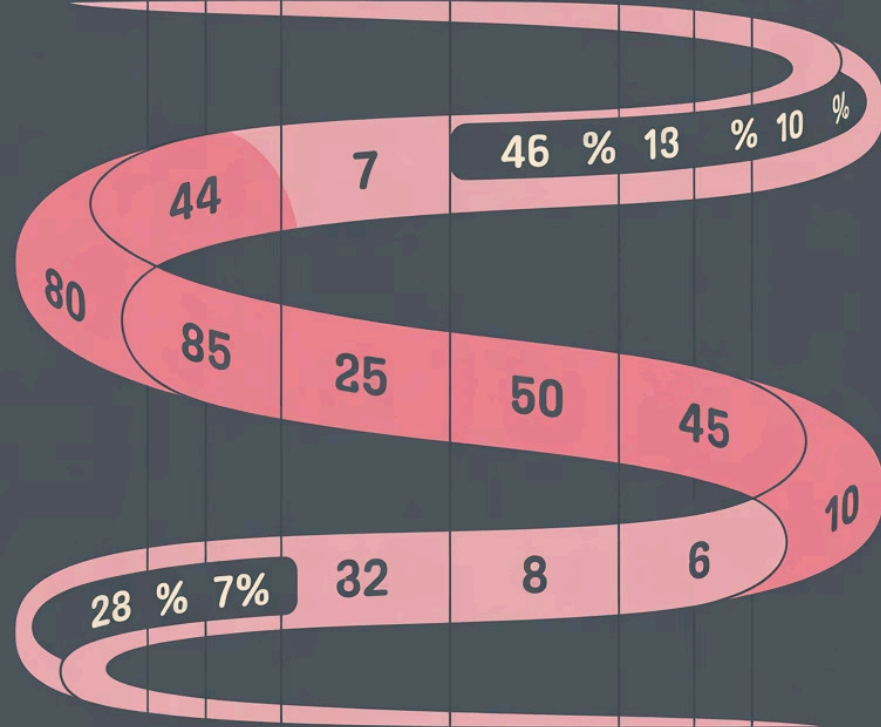
3

Calculate Difference

`=ABS(A2 - B2)`

Always returns positive result

122330



156789

EVEN & ODD: Rounding Functions



EVEN Function

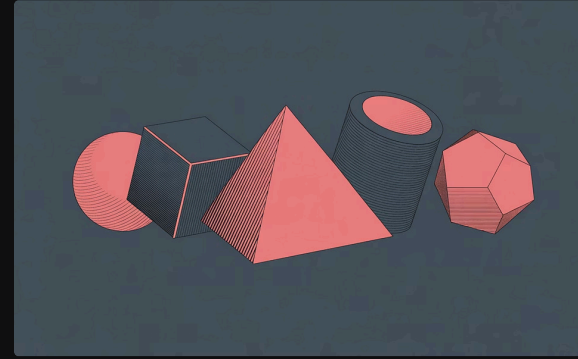
Rounds a number *up* to the nearest even integer. Useful for packing logic where items must be sold in pairs.

Round to Even

=EVEN(3) → 4

=EVEN(-3) → -4

Rounds away from zero



ODD Function

Rounds a number *up* to the nearest odd integer. Helpful for creating specific number sequences and patterns.

Round to Odd

=ODD(4) → 5

=ODD(2) → 3

Always rounds up



QUOTIENT: Integer Division

What It Returns

Returns only the integer portion of a division, discarding any remainder. Perfect for "how many full groups" questions.

Common Applications

- Team formation calculations
- Inventory packaging
- Batch processing
- Resource allocation

01

Basic Division

=QUOTIENT(10, 3) → **3**

3 goes into 10 three times

03

Syntax Structure

=QUOTIENT(numerator, denominator)

Real-World Example

"How many full teams of 5 can we make from 23 people?"

=QUOTIENT(23, 5) → **4 teams**

The remainder (3 people) is ignored.

02

Inventory Calculation

=QUOTIENT(100, 12) → **8**

Full dozens in 100 items

POWER & SQRT: Exponents and Roots



POWER Function

Returns a number raised to a specified power. Used in engineering, statistics, and geometric calculations like area of circles or compound interest.

Syntax: =POWER(number, power)

Power Example

=POWER(5, 2) → 25

5 squared equals 25



SQRT Function

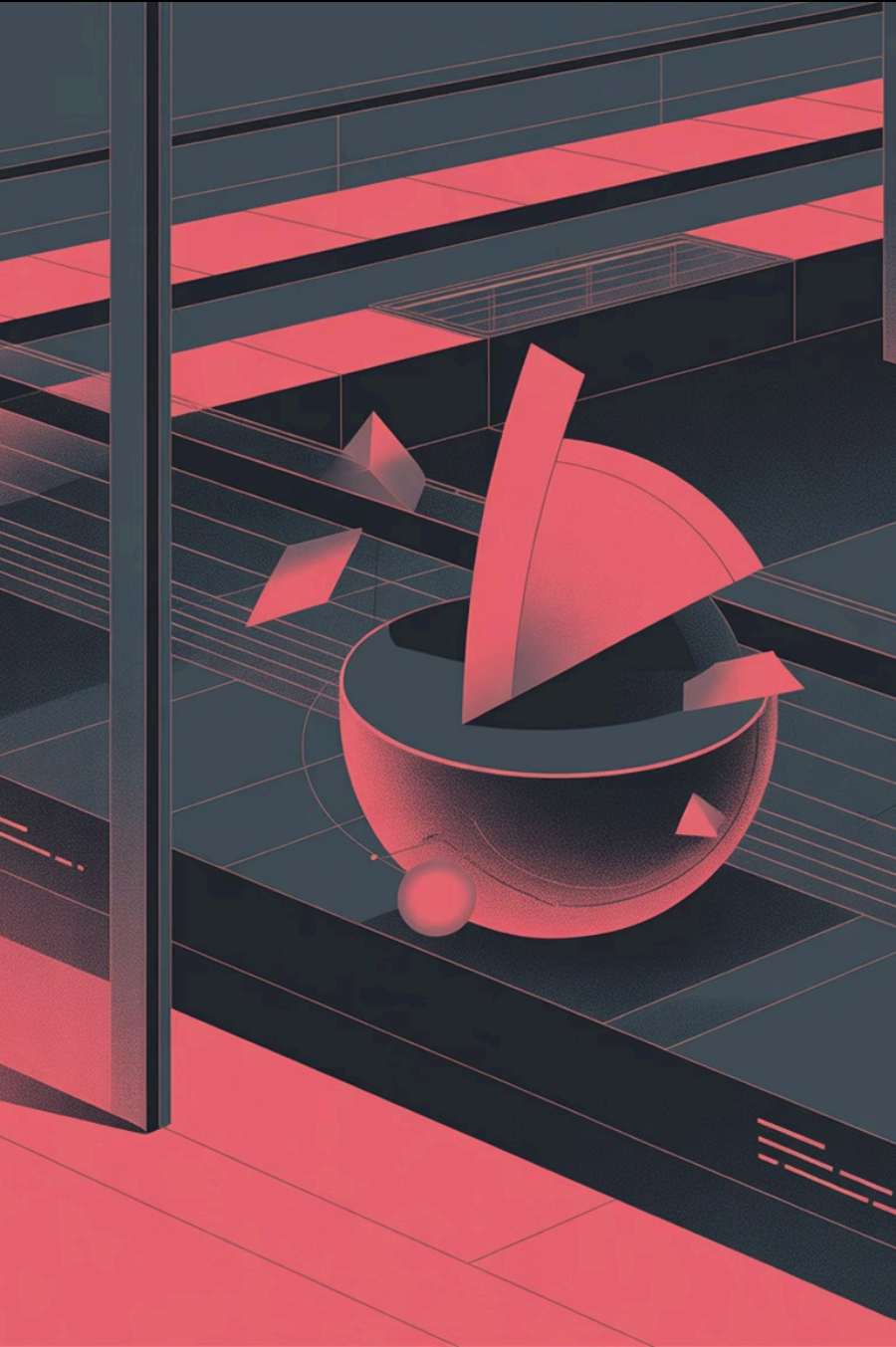
Returns the square root of a number. Essential for distance calculations, standard deviation, and solving quadratic equations.

Syntax: =SQRT(number)

Square Root Example

=SQRT(49) → 7

The square root of 49 is 7



MOD: Finding Remainders

The Complement to QUOTIENT

Returns the remainder after division. While QUOTIENT tells you how many full times, MOD tells you what's left over.

Essential Applications

- Checking if numbers are even or odd
- Cycle detection
- Data validation
- Pattern recognition

Find Remainder

=MOD(10, 3) → 1

10 divided by 3 equals 3 with remainder 1

Check Even Numbers

=MOD(A2, 2)

If result is 0, the number is Even

Syntax

=MOD(number, divisor)

FACT: Factorial Calculations

What Is a Factorial?

Returns the product of all positive integers from 1 up to the given number.
Written as $n!$

Example: $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$

Real-World Use

Mostly used in probability and statistics for permutations and combinations.

"In how many ways can 5 books be arranged on a shelf?"

Answer: $=\text{FACT}(5) \rightarrow 120$ arrangements

5

Input Number

The value you want to calculate factorial for

120

Result

$=\text{FACT}(5)$ returns 120

1

Base Case

$\text{FACT}(0)$ and $\text{FACT}(1)$ both return 1



An illustration of several dice of various sizes and orientations scattered on a shelf. The dice are rendered in a stylized, isometric manner with some faces showing numbers. The background consists of vertical lines and a warm, orange-toned light source from the left, creating a sense of depth and perspective.

RAND: Generating Random Numbers

What It Does

Returns a random decimal number greater than or equal to 0 and less than 1. The value changes every time your worksheet recalculates.

Common Applications

- Monte Carlo simulations
- Random sampling
- Game development
- Drawing random winners
- Generating test data

Basic Random

`=RAND()`

Returns a random decimal between 0 and 1 (e.g., 0.8745, varies each calculation)

Random 0-100

`=RAND()*100`

Generates a random number between 0 and 100

Fixed Random

Copy and paste as values to freeze the random number

 **Pro Tip:** Press F9 to recalculate and generate new random numbers instantly!